

# Compact Vibrating Fork Level Switch

**For Liquid Applications Model: VFSL-700L**



## PRODUCT DESCRIPTION:

The VFSL-700L Compact Vibrating Fork Type Liquid Level Switch is a robust and reliable point level detection instrument designed for accurate monitoring of liquid levels in tanks, vessels and process equipment. The instrument operates on the proven vibrating fork principle and provides dependable detection of High, Mid or Low liquid levels. The compact design incorporates integral electronics, adjustable sensitivity, adjustable time delay, LED status indication and selectable fail-safe operation, enabling reliable performance and simple commissioning. Its rugged construction makes it suitable for a wide range of demanding industrial applications where dependable point level detection is required.

## SALIENT FEATURES:

- Wide Range of Liquid Level Applications
- Compact, Rugged and Reliable Construction
- Easy Installation, Commissioning and Maintenance
- Weatherproof / Flameproof Enclosure available as per requirement
- SS 316 Vibrating Fork for reliable operation in harsh process environments
- Adjustable ON/OFF Time Delay for flexible process control
- Field-Configurable Fail-Safe High / Low Selection
- Adjustable Sensitivity to suit different liquid media
- Integral Electronics for simple installation and operation
- LED Status Indication for easy monitoring
- No Moving Parts for low maintenance and long service life
- Suitable for Conductive and Non-Conductive Liquids
- Reliable High, Mid and Low-Level Detection
- Suitable for Low to High Viscosity Liquids
- Fast Response and Stable Level Detection

## LIQUID LEVEL APPLICATIONS:

Water, Oils, Chemicals, Acids, Alkalis, Paints, Food & Beverages, Pharmaceuticals, Dairy Products, Detergents, Adhesives, Fertilizers, Slurries and other Low to High Viscosity Liquids.

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LEVTRON  
SENSING INSTRUMENTS

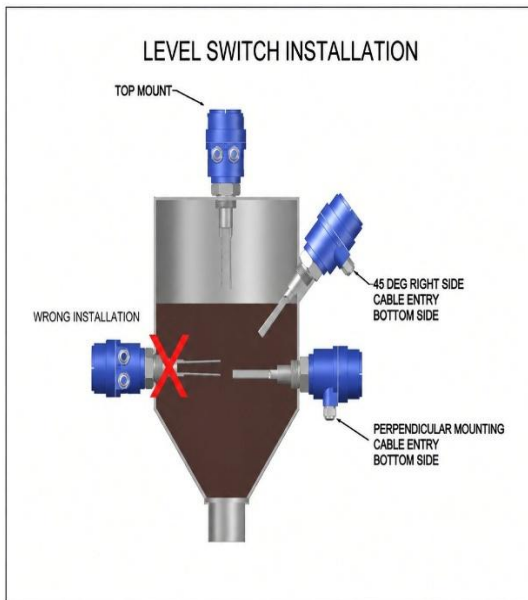


### OPERATING PRINCIPLE:

The tuning fork is continuously vibrated at its mechanical resonant frequency by piezoelectric elements. When the liquid comes into contact with the fork tines, the vibration is damped due to the resistance and mass loading of the liquid. This change in vibration is electronically detected and processed by the internal electronics to generate a reliable level switching signal. When the liquid level falls below the fork, the tines resume their normal vibration, and the switch returns to its normal operating state. The output relay provides a High- or Low-Level switching signal and can be used for level alarm and process control applications such as pump control, dry-run protection, overflow protection and valve control.

The VFLS-700L provides reliable point-level detection for a wide range of liquids, with no moving mechanical parts in the sensing mechanism, ensuring dependable operation and low maintenance.

### PROBE CONSTRUCTION SUITABLE FOR LIQUID APPLICATIONS:



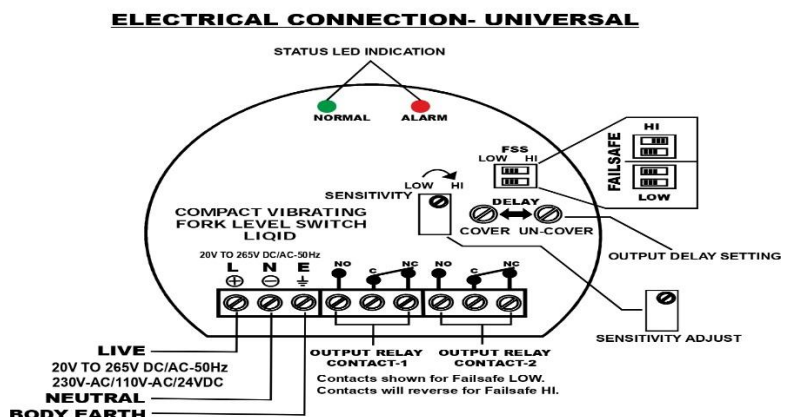
The VFLS-700L Liquid Vibrating Fork Level Switch is designed for reliable point-level detection in a wide range of liquids such as water, oils, resins, acids and chemicals. The robust SS 316 vibrating fork provides reliable operation under demanding process conditions. When the liquid level reaches the vibrating fork, the fork vibration is damped by the surrounding liquid. This change in vibration is detected by the internal electronics and processed to generate a reliable switching signal to the integrated relay output. The built-in timer function provides adjustable ON/OFF time delay for stable level detection and helps minimize unwanted switching. The vibrating fork operates at a nominal frequency of approximately 350 Hz ( $\pm 10$  Hz), providing consistent and dependable point-level detection.

### COMPONENTS OF COMPACT LEVEL SWITCH ARE EXPLAINED IN FIGURE



- ----Aluminum Housing
- ----Cable Entry
- -----Mounting Thread
- -----Extension pipe
- ----- Tuning Fork

### ELECTRICAL CONNECTION UNIVERSAL /230 VAC/110VAC/24 VDC



# Compact Vibrating Fork Level Switch

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### DATA SHEET

#### VFLS-700 Vibrating Fork Level Switch Selection / Ordering Information

| Selection              | Code                      | Description                                                |
|------------------------|---------------------------|------------------------------------------------------------|
| Model No.              | VFLS-700                  | Vibrating Fork Level Switch                                |
| Application            | L                         | Liquid                                                     |
| Mounting Type          | IN/RE                     | Integral / Remote                                          |
| Probe Type             | RD / RP                   | Rigid Pipe / Hose Flexible                                 |
| Insulation Coating     | NI/PT                     | Non- Coating /PTFE Coating                                 |
| Probe Diameter         | D1                        | Standard 30 mm                                             |
| Probe Material         | 04 /4L/ 16/ 6L/MS/ SM     | SS304/SS304L / SS316/ SS316L /Mild steel/ Special material |
| Power Supply           | AC/DC/A1/UN               | 230VAC / 24VDC / 110VAC / Universal                        |
| Output                 | DP/SP/PN                  | DPDT Relay /SPDT Relay / PNP Relay                         |
| Switching Delay        | ND / DA                   | No Delay / Adjustable                                      |
| Indication             | L                         | LED                                                        |
| Enclosure              | WP / FL                   | Weatherproof / Flameproof                                  |
| Connection             | T                         | Threaded                                                   |
|                        | F                         | Flanged                                                    |
| Mounting               | B1 / B2 / B3/B4           | 1" / 1½" / 2" BSP/ Special Mounting                        |
|                        | N1 / N2 / N3/N4           | 1" / 1½" / 2" NPT/ Special Mounting                        |
| Flange Rating          | A1 / A3 / A6              | ANSI 150# / ANSI 300# /ANSI 600#                           |
| Flange Facing          | RF / FF                   | Raised Face / Flat Face                                    |
| MOC-Process Connection | S4 / S6/ S16/MS/SL        | SS304 / SS316 /SS316L /Mild steel / Special material       |
| Flange Size            | 01 / 15 / 02 / 03 / 04/PS | 1" / 1½" / 2" / 3" / 4"/ Special Process                   |
| Pressure               | P1 /P2/P3/P4              | Standard 0 to 2KG/ 2 To 10kg /10kg to 40kg / Customized    |
| Temperature            | T1                        | Max –150°C                                                 |
| Remote Cable           | 05 / SR                   | 5 /Special Customer Requirement                            |
| Cable Gland            | G1/ G2 / G3               | PG13.5 / ½"NPT / ¾" ET                                     |
| Material Certificate   | M0 / MC                   | Not Required / Material test certificate                   |

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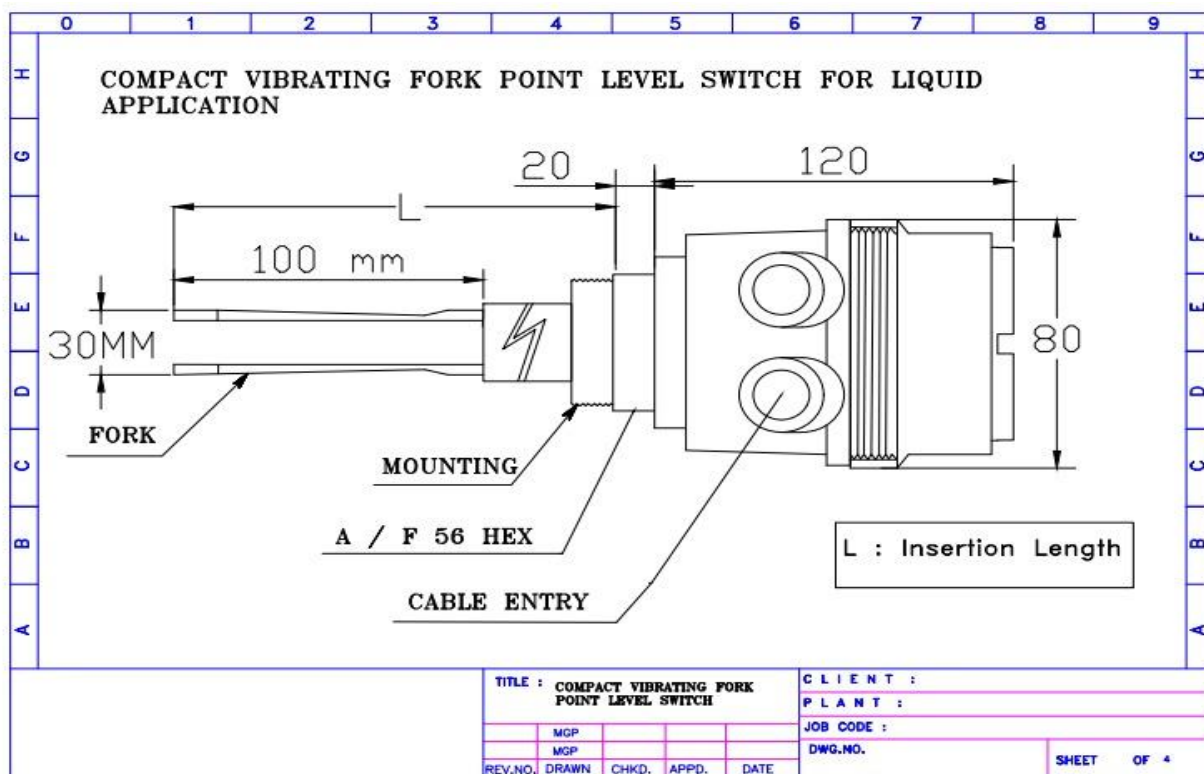
## For Liquid Applications Model: VFLS-700L

| Selection        | Code         | Description                                |
|------------------|--------------|--------------------------------------------|
| Test Certificate | TC / CC      | Test Certificate / Calibration Certificate |
| Documentation    | DS / GA / IR | Datasheet / GA / Inspection Report         |
| Tagging          | TAG          | Customer Tag No.                           |
| Probe Length     | XXXX         | 150 mm TO Customized                       |

### INFORMATION REQUIRED Before Purchase

- Process Parameters: Process Connection, Process Temperature, Vessel pressure if any. Application, characteristics: Density, Viscosity, Sp. Gr.
- Supply Voltage, Probe length, Mounting etc.

### Mechanical Drawing:



Manufactured By:

**Levtron Instruments Pvt Ltd.**

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